

enormously and at the same time with modern CPU manufacturing a lot of the electrical power constraints have fallen. Today, you wouldn't have a Kinect Controller for the Xbox 360 if it weren't for embedded systems!

The rise of embedded systems?

Due to systems-on-a-chip integrations which combine a microprocessor, RAM, I/O controllers and other control systems into one integrated circuit package, embedded systems were able to emerge into the marketplace in the 1980s. They were, of course, less flexible than conventional computers but they were also much cheaper. Unfortunately, due to power utilisation, electrical, and timing constraints, most of the embedded systems remained as 8-bit systems while the rest of the computing world moved on to 16-bit and 32-bit processing.

So now processors have shrunk in size along with an increase in performance; power consumption has been drastically reduced, and the cost of processors has fallen considerably to quite an affordable level. In addition, the design cycle and total development time has been reduced because of two reasons; there is now more awareness that rather than using a totally hardwired electronic system, if a programmable processor is incorporated in the circuit it makes the design more robust; and the concept of a development environment where the system can be prototyped and a simulation can also be carried out. Along with this, the development of Java and other standard run-time platforms, the emergence of integrated software environments which simplified development of the applications, and debatably the biggest of them all is the coming together of Embedded systems and the internet has contributed greatly to the rise of embedded systems. This makes the networking of many Embedded systems to operate as a part a system across networks like LAN, WAN, or the internet possible.